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CYCLE SHEET– III (PL/SQL Exercises)

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SUBJECT CODE : ISWE103P

SUBJECT LAB : DATABASE SYSTEM LAB

LAB SLOT : L33 +L34

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1. Write a PL/SQL block to display the reverse of numbers between 1 and 100.

```
SQL> SET SERVEROUTPUT ON;
SQL>  DECLARE
  2  i NUMBER;
  3  BEGIN
  4  FOR i IN REVERSE 1..100 LOOP
  5  DBMS_OUTPUT.PUT_LINE(i);
  6  END LOOP;
  7  END;
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PL/SQL procedure successfully completed.
```

2. Project (Proj_num, Name, Location, Controlling Department, Budget)

Write a PL/SQL program to display the project name which has the highest budget than others.

```
SQL> CREATE TABLE Project(Proj_num INT,Name VARCHAR(20),Location VARCHAR(30),Controlling_Department INT, Budget INT);

Table created.
```

```
SQL> SET SERVEROUTPUT ON;
SQL> DECLARE
  2   v_name    Project.Name%TYPE;
  3  v_budget  Project.Budget%TYPE;
  4  BEGIN
  5  SELECT Name, Budget
  6  INTO v_name, v_budget
  7  FROM Project
  8  WHERE Budget = (SELECT MAX(Budget) FROM Project);
  9
 10  DBMS_OUTPUT.PUT_LINE('Project with Highest Budget:');
 11  DBMS_OUTPUT.PUT_LINE('Name   : ' || v_name);
 12   DBMS_OUTPUT.PUT_LINE('Budget : ' || v_budget);
 13  END;
 14  /
Project with Highest Budget:
Name   : PRJ2024
Budget : 750000

PL/SQL procedure successfully completed.
```

3. Employee (SSN, Name (fname, Minit, Lname), Sex, Address, salary, DOB, Department, designation, SupervisorSSN) .

Write a PL/SQL program to get the employee number and check that the employee has got salary more than 50000. Otherwise, print the suitable message to denote that the employee is getting salary less than 50000.

```
SQL> SET SERVEROUTPUT ON;
SQL>  SET DEFINE ON;
SQL>  DECLARE
  2  v_ssn    Emp_1.SSN%TYPE := &emp_ssn;
  3  v_salary Emp_1.salary%TYPE;
  4  v_fname  Emp_1.fname%TYPE;
  5  v_lname  Emp_1.Lname%TYPE;
  6  BEGIN
  7  SELECT fname, Lname, salary INTO v_fname, v_lname, v_salary FROM Emp_1 WHERE SSN = v_ssn;
  8  DBMS_OUTPUT.PUT_LINE('Employee No   : ' || v_ssn);
  9  DBMS_OUTPUT.PUT_LINE('Employee Name : ' || v_fname || ' ' || v_lname);
 10  DBMS_OUTPUT.PUT_LINE('Salary        : ' || v_salary);
 11
 12  IF v_salary > 50000 THEN
 13   DBMS_OUTPUT.PUT_LINE('Status : Employee salary is more than 50000.');
 14  ELSE
 15  DBMS_OUTPUT.PUT_LINE('Status : Employee is getting salary less than 50000.');
 16  END IF;
 17  EXCEPTION
 18  WHEN NO_DATA_FOUND THEN
 19  DBMS_OUTPUT.PUT_LINE('No employee found with SSN: ' || v_ssn);
 20  WHEN OTHERS THEN
 21  DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
 22  END;
 23  /
Enter value for emp_ssn: 101
old   2: v_ssn    Emp_1.SSN%TYPE := &emp_ssn;
new   2: v_ssn    Emp_1.SSN%TYPE := 101;
Employee No   : 101
Employee Name : Raj Kumar
Salary        : 60000
Status : Employee salary is more than 50000.

PL/SQL procedure successfully completed.
```

```
SQL> /
Enter value for emp_ssn: 103
old   2: v_ssn    Emp_1.SSN%TYPE := &emp_ssn;
new   2: v_ssn    Emp_1.SSN%TYPE := 103;
Employee No   : 103
Employee Name : Arjun Singh
Salary        : 55000
Status : Employee salary is more than 50000.

PL/SQL procedure successfully completed.

SQL> /
Enter value for emp_ssn: 190
old   2: v_ssn    Emp_1.SSN%TYPE := &emp_ssn;
new   2: v_ssn    Emp_1.SSN%TYPE := 190;
No employee found with SSN: 190

PL/SQL procedure successfully completed.

SQL> /
Enter value for emp_ssn: 102
old   2: v_ssn    Emp_1.SSN%TYPE := &emp_ssn;
new   2: v_ssn    Emp_1.SSN%TYPE := 102;
Employee No   : 102
Employee Name : Priya Sharma
Salary        : 45000
Status : Employee is getting salary less than 50000.

PL/SQL procedure successfully completed.
```

4. Write a procedure to display no of employee working in a department “SALES”;

```
SQL> SET SERVEROUTPUT ON;
SQL> CREATE OR REPLACE PROCEDURE count_sales_emp IS v_count NUMBER;
  2  BEGIN
  3  SELECT COUNT(*)
  4  INTO v_count FROM Emp_1 WHERE Department = 'SALES';
  5  DBMS_OUTPUT.PUT_LINE('Number of employees in SALES department: ' || v_count);
  6  END;
  7  /

Procedure created.

SQL> BEGIN
  2  count_sales_emp;
  3  END;
  4  /
Number of employees in SALES department: 2

PL/SQL procedure successfully completed.
```

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5. Write a program in PL/SQL to retrieve the records from the employees table and display them using cursors.

```
SQL> SET SERVEROUTPUT ON;
SQL> DECLARE
  2   CURSOR emp_cursor IS  SELECT SSN, fname, Minit, Lname, Sex, Address, salary, DOB, Department, designation, SupervisorSSN  FROM Emp_1;
  3   v_ssn     Emp_1.SSN%TYPE;
  4  v_fname   Emp_1.fname%TYPE;
  5  v_minit   Emp_1.Minit%TYPE;
  6  v_lname   Emp_1.Lname%TYPE;
  7   v_sex     Emp_1.Sex%TYPE;v_address Emp_1.Address%TYPE;v_salary  Emp_1.salary%TYPE;
  8   v_dob     Emp_1.DOB%TYPE; v_dept    Emp_1.Department%TYPE;v_desig   Emp_1.designation%TYPE;
  9  v_supssn  Emp_1.SupervisorSSN%TYPE; v_count   NUMBER := 0;
 10  BEGIN
 11  OPEN emp_cursor;
 12  DBMS_OUTPUT.PUT_LINE('-------- Employee Records --------');
 13  LOOP
 14  FETCH emp_cursor INTO v_ssn, v_fname, v_minit, v_lname, v_sex, v_address, v_salary, v_dob, v_dept, v_desig, v_supssn;
 15  EXIT WHEN emp_cursor%NOTFOUND;
 16  v_count := v_count + 1;
 17  DBMS_OUTPUT.PUT_LINE('SSN : ' || v_ssn);
 18  DBMS_OUTPUT.PUT_LINE('Name  :'||v_fname || ' ' || v_minit || ' '||v_lname);
 19  DBMS_OUTPUT.PUT_LINE('Sex  : ' || v_sex);
 20  DBMS_OUTPUT.PUT_LINE('Address  : ' || v_address);
 21  DBMS_OUTPUT.PUT_LINE('Salary : ' || v_salary);
 22  DBMS_OUTPUT.PUT_LINE('DOB : ' || v_dob);
 23  DBMS_OUTPUT.PUT_LINE('Department  : ' || v_dept);
 24  DBMS_OUTPUT.PUT_LINE('Designation : ' || v_desig);
 25  DBMS_OUTPUT.PUT_LINE('Supervisor  : ' || v_supssn);
 26  DBMS_OUTPUT.PUT_LINE('--------------------------------');
 27  END LOOP;
 28  CLOSE emp_cursor;
 29  DBMS_OUTPUT.PUT_LINE('Total Records: ' || v_count);
 30  END;
 31  /
-------- Employee Records --------
```

```
-------- Employee Records --------
SSN : 102
Name  :Priya S Sharma
Sex  : F
Address  : Mumbai
Salary : 45000
DOB : 22-AUG-95
Department  : HR
Designation : Analyst
Supervisor  : 101
---------------------------------
SSN : 103
Name  :Arjun R Singh
Sex  : M
Address  : Delhi
Salary : 55000
DOB : 01-DEC-88
Department  : SALES
Designation : Engineer
Supervisor  : 101
---------------------------------
SSN : 148
Name  :Arun R Iyer
Sex  : M
Address  : Delhi
Salary : 45000
DOB : 11-DEC-08
Department  : HR
Designation : Engineer
Supervisor  : 109
---------------------------------
SSN : 122
Name  :Priya S Sharma
Sex  : F
Address  : Mumbai
Salary : 45000
DOB : 22-AUG-95
Department  : HR
Designation : Analyst
Supervisor  : 108
---------------------------------
```

```
---------------------------------
SSN : 122
Name  :Priya S Sharma
Sex  : F
Address  : Mumbai
Salary : 45000
DOB : 22-AUG-95
Department  : HR
Designation : Analyst
Supervisor  : 108
---------------------------------
SSN : 123
Name  :Arjun R Singh
Sex  : M
Address  : Delhi
Salary : 55000
DOB : 01-DEC-88
Department  : SALES
Designation : Engineer
Supervisor  : 101
---------------------------------
SSN : 134
Name  :Meena T Nair
Sex  : F
Address  : Chennai
Salary : 48000
DOB : 15-MAR-92
Department  : SALES
Designation : Executive
Supervisor  : 105
---------------------------------
SSN : 165
Name  :Vikram P Reddy
Sex  : M
Address  : Hyderabad
Salary : 72000
DOB : 30-JUL-85
Department  : IT
Designation : Senior Engineer
Supervisor  : 106
---------------------------------
```

```
SSN : 165
Name  :Vikram P Reddy
Sex  : M
Address  : Hyderabad
Salary : 72000
DOB : 30-JUL-85
Department  : IT
Designation : Senior Engineer
Supervisor  : 106
---------------------------------
SSN : 106
Name  :Sneha A Iyer
Sex  : F
Address  : Bangalore
Salary : 39000
DOB : 10-NOV-98
Department  : SALES
Designation : Trainee
Supervisor  : 103
---------------------------------
Total Records: 8

PL/SQL procedure successfully completed.
```

6. Create a trigger named update_restrict which display the warning message as "Deletion Not Permitted" if any one tries to delete a record from an employee table.

```
SQL> CREATE OR REPLACE TRIGGER update_restrict
  2  BEFORE DELETE ON Emp_1
  3  FOR EACH ROW
  4  BEGIN
  5  RAISE_APPLICATION_ERROR(-20001, 'Deletion Not Permitted');
  6  END;
  7  /

Trigger created.

SQL> DELETE FROM Emp_1 WHERE SSN = 101;
DELETE FROM Emp_1 WHERE SSN = 101
            *
ERROR at line 1:
ORA-20001: Deletion Not Permitted
ORA-06512: at "STUDENT.UPDATE_RESTRICT", line 2
ORA-04088: error during execution of trigger 'STUDENT.UPDATE_RESTRICT'
```

```
SQL> SELECT trigger_name, status FROM user_triggers WHERE table_name = 'Emp_1';

no rows selected

SQL>  SELECT trigger_name, status FROM user_triggers WHERE table_name = 'EMP_1';

TRIGGER_NAME                   STATUS
------------------------------ --------
UPDATE_RESTRICT                ENABLED

SQL> SELECT trigger_name, status FROM user_triggers WHERE table_name = 'EMPLOYEE';

no rows selected
```